



EXAM PAPERS PRACTICE

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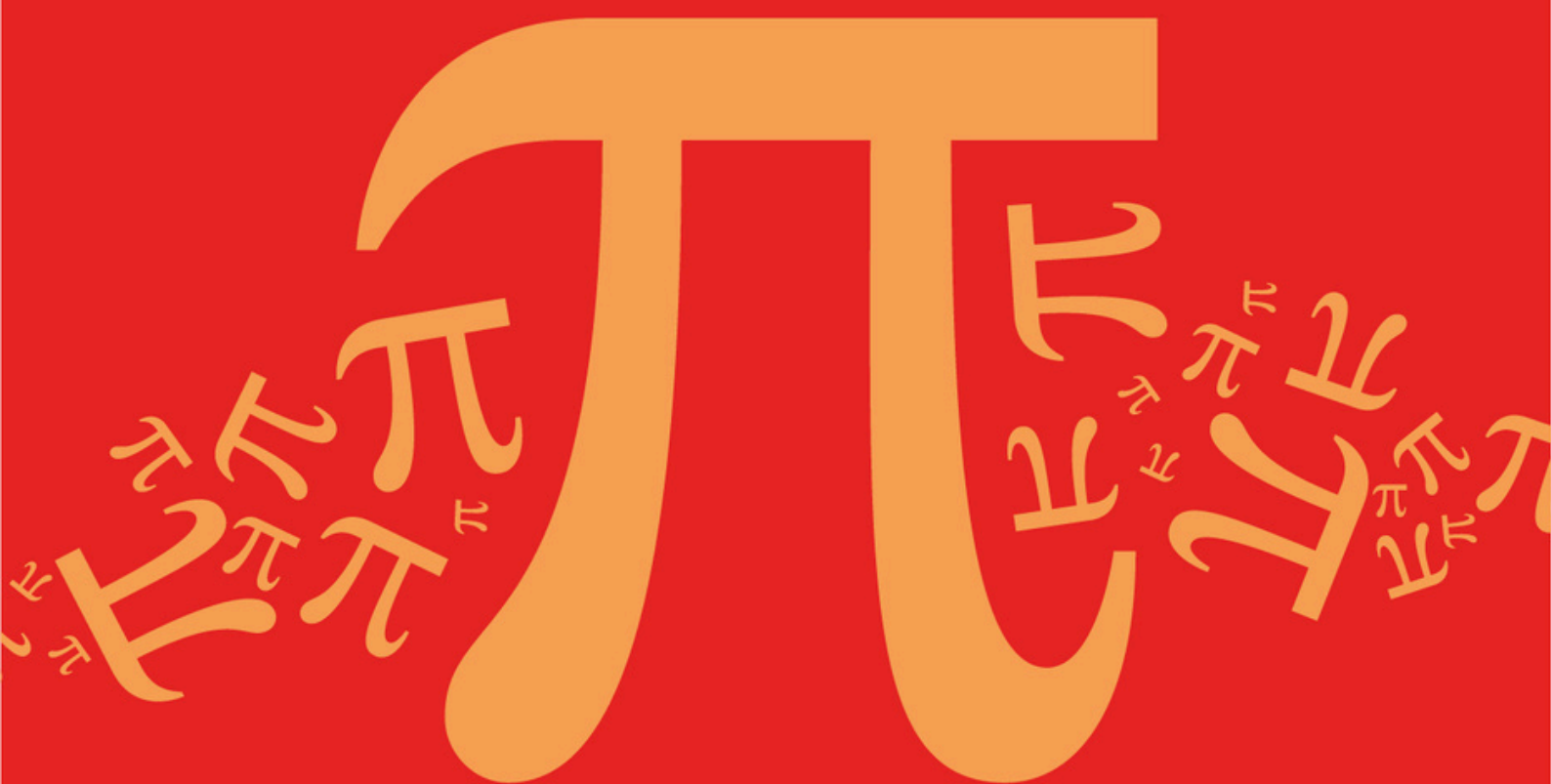
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

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AQA A Level Further Mathematics (7367)



Topic	B: Complex Numbers
Sub topic	B7: Loci

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic

B: Complex Numbers

Sub topic

B7: Loci

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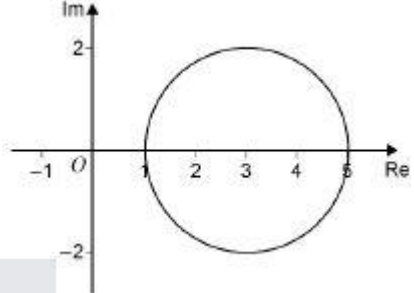
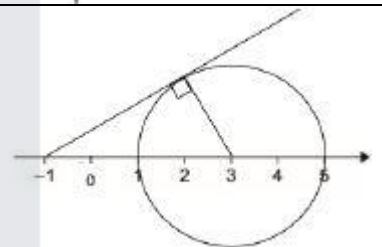
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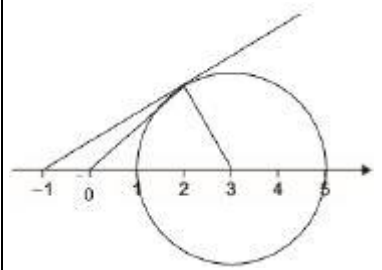
**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

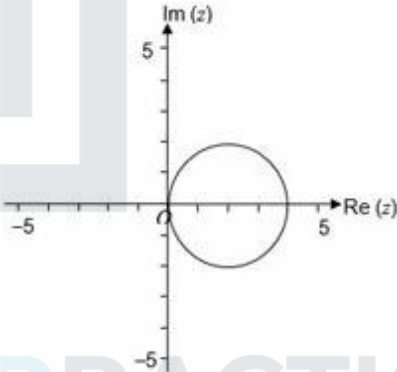
Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Draws a circle with centre (3,0) and radius 2. Accept freehand circle. Ignore any straight lines drawn on the diagram.	AO1.1b	B1	
(b) (i)	Uses fully correct method for $\sin \alpha$ or $\cos \alpha$ or $\tan \alpha$ $\sin \alpha = \frac{2}{4} \quad \cos \alpha = \frac{\sqrt{4^2 - 2^2}}{4}$ $\tan \alpha = \frac{2}{\sqrt{4^2 - 2^2}}$	AO3.1a	M1	
	Obtains correct value for α Accept 0.52(35987756) Condone 30°	AO1.1b	A1	$\sin \alpha = \frac{2}{4}$ $\alpha = \sin^{-1}\left(\frac{2}{4}\right)$ $\alpha = \frac{\pi}{6}$

	Marking Instructions	AO	Marks	Typical Solution
(ii)	Forms an equation in r using cosine rule or equivalent. Follow through their α . Or forms a correct equation in x and y .	AO3.1a	M1	 $r^2 = 2^2 + 3^2 - 2 \times 2 \times 3 \times \cos \frac{\pi}{3}$ $r = \sqrt{7}$
	Forms an equation in θ using sine rule or equivalent. Follow through their α . Or forms a second correct equation in x and y .	AO1.1a	M1	$\frac{\sin \theta}{2} = \frac{\sin \frac{\pi}{3}}{\sqrt{7}}$
	Obtains correct value for r or θ .	AO1.1b	A1	$\sin \theta = 2 \times \frac{\sqrt{3}}{2} \div \sqrt{7} = \frac{\sqrt{21}}{7}$

Or obtains correct values for x and y.			$\theta = 0.71$
Expresses w in the required form. Accept 2.6[45751311] or $\sqrt{7}$ for r Accept 0.71[37243789] or $\sin^{-1}\left(\frac{\sqrt{21}}{7}\right)$ or $\sin^{-1}\left(\sqrt{\frac{3}{7}}\right)$ for θ	AO1.1b	A1	$w = 2.6(\cos 0.71 + i \sin 0.71)$
Total 7 marks			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Draws 'circle' with centre $2 + 0i$ Ignore other features	AO1.1a	M1	
	Draws circle passing through $(0, 0)$, $(4, 0)$, close to $(2, 2)$ and $(2, -2)$ with Imaginary axis tangential	AO1.1b	A1	
(b)	Uses mod/arg forms	AO3.1a	M1	$z - 2 = 2 \left[\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right) \right]$
	Substitutes exact values for cos and sin Allow one slip	AO1.1a	M1	$= 2 \left[\frac{1}{2} + i \left(-\frac{\sqrt{3}}{2} \right) \right]$
	Obtains result in exact form	AO1.1b	A1	$z = 3 - \sqrt{3}i$
				Total 5 marks

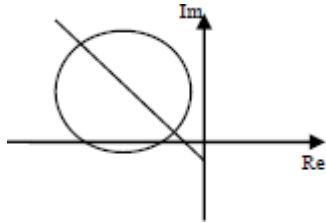
Q3.

(a) $|4 - 4i| = \sqrt{16+16} = \sqrt{32} = 4\sqrt{2}$
 verification that $|-2+i+6-5i| = 4\sqrt{2}$

B1

$\arg(-2 + 2i) = \pi - \tan^{-1}(1) = \frac{3\pi}{4}$
 verification that $\arg(z + i) = \frac{3\pi}{4}$

B1



2

(b) Circle

freehand circle sketched

M1

Centre at $-6 + 5i$

clear from diagram or centre stated

A1

Cutting Re axis but not cutting Im axis

A1

"Straight" line

freehand line

M1

Half line from $0 - i$

not horizontal or vertical but end point at $0 - i$ must be clear from diagram / stated

A1

gradient -1 (approx)

making 45° to negative Re axis and positive Im axis

A1

6

(c) Calculation based on fact that L_2 passes through centre of L_1

idea of vector $\begin{bmatrix} -4 \\ 4 \end{bmatrix}$ from centre

M1

Q represents $-10 + 9i$

must write as a complex number

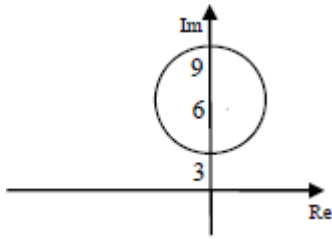
A1

2

[10]

Q4.

(a)



Circle

freehand circle

M1

Centre at 6i

6 marked on Im axis as centre

A1

Radius 3 & cutting positive Im axis twice

radius of 3 clearly indicated with circle in position shown

A1

3

(b) (i) (Max $|z|$ is) 9
1

B1

1

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(ii) Tangent from O to circle

FT their circle position

M1

Angle of $\frac{\pi}{6}$ or $\frac{\pi}{3}$ correctly marked

PI; condone degrees for first A1

A1

(Max $\arg z$ is) $\frac{2\pi}{3}$
exactly this

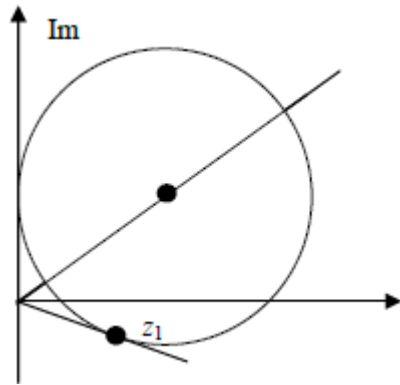
A1cso

3

[7]

Q5.

(a)



Half-line with gradient < 1

condone a short line, ie it stops at or inside circle

B1

1

- (b) (i) Circle centre on L , x-coord 6 indicated

B1

touching $\text{Re } z = 0$ not at $(0, 0)$

not touching Re axis

B1

2

- (ii) y-coord of centre is $2\sqrt{3}$ or $\frac{6}{\sqrt{3}}$
OE; PI

B1

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$$z_0 = 6 + 2\sqrt{3}i,$$

ft error in coords of centre

B1F,

$$k = 6$$

B1

- (iii) Point z_1 shown

PI

B1

$$\arg \pi_1 = -\frac{1}{6}$$

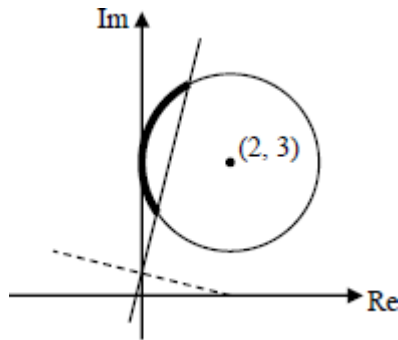
B1

2

[8]

Q6.

(a)



(i) Circle

Convex loop

Correct centre

Some indication of position of centre

Touching Im axis

B1

B1

B1

3

(ii) Straight line well to left of centre

$\frac{1}{2}$ line through $(0, \frac{1}{2})$ B0

through $(0, \frac{1}{2})$

Point approximately between 0 and 1

B1

$\perp$ to line joining $(-2, 1)$ and $(2, 0)$

B1

NB

0 / 3 for line parallel to x-axis

0 / 3 for line joining the two points $(-2, 1)$ and $(2, 0)$

0 / 3 for line joining $(0, 0)$ to centre of circle

3

(b) Minor arc indicated

ft incorrect position of line or circle

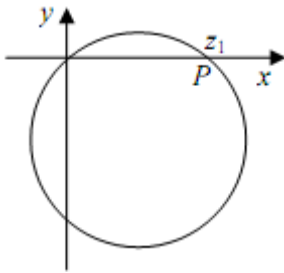
B1F

1

[7]

Q7.

(a)



Circle

B1

correct centre

B1

through (0, 0)

B1

3

- (b) (i) z_1 correctly chosen
ft if circle encloses (0, 0)

B1F

1

- (ii) $|z_1| = 8$
ft if centre misplotted

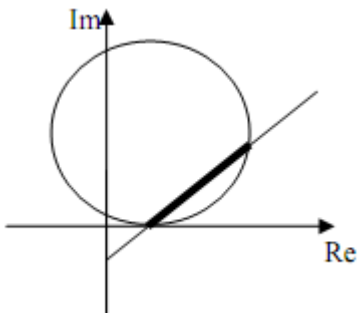
B1F

1

[5]

Q8.

(a)



Use average of whole question if 2 diagrams used

- (i) Circle
Circle in any position

B1

correct centre
Must be shown

B1

touching x -axis
ft incorrect centre

B1F

3

(ii) half-line

B1

through $(0, -2)$
Can be inferred

B1

through point of contact of circle with
 x -axis

B1

3

(b) Inside circle

B1

On line
ft errors in position of line and circle

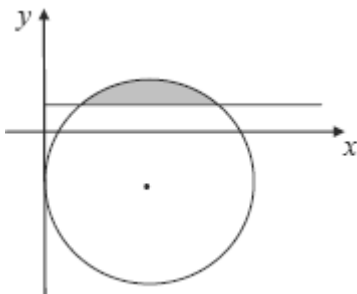
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B1F

2

[8]

Q9.



(a) (i) Circle

B1

Correct centre
 x -coordinate $\approx -2 \times y$ -coordinate in

correct quadrant; condone $(4, -2i)$

B1

Touching y-axis

B1

3

(ii) Straight line

B1

parallel to x-axis

B1

through $(0, 1)$

Assume $(0, 1)$ if distance up y-axis is half distance to top of circle; no other shading outside circle

B1

3

(b) Shading: inside circle

B1F

above line

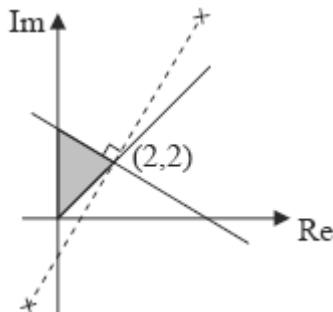
B1F

2

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 Whole question reflected in x-axis loses 2 marks

[8]

Q10.



(a) $|2 + 2i + 1 + 3i| = |2 + 2i - 5 - 7i|$

Clearly shown do not allow

$|3 + 5i| = |-3 - 5i|$ without comment

B1

$$\arg(2 + 2i) = \frac{\pi}{4}$$

Clearly shown

B1

2

- (b) L_1 : straight line with negative gradient

B1

perpendicular to line joining
 $(-1, -3)$ to $(5, 7)$

B1

through $(2, 2)$

*The point $(2, 2)$ must be shown either by
 $(2, 2)$ or $2 + 2i$ or with numbered axes*

B1

L_2 : half line through O

B1

through $(2, 2)$

B1

5

- (c) Shading between $\frac{\pi}{4}$ and $\frac{\pi}{2}$

No marks for shading if circles drawn in (b)

B1

Below L_1

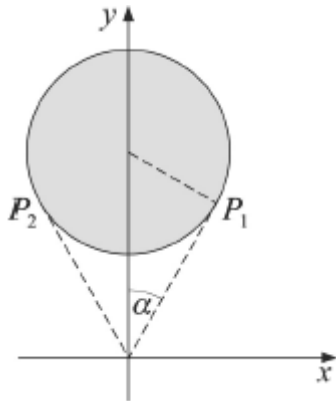
B1

2

[9]

Q11.

(a)



Circle

B1

Correct centre

B1

Correct radius

B1

Inside shading

B1F

4

- (b) Correct points P_1 and P_2 indicated

Possibly by tangents drawn
ft mirror image of circle in x-axis

B1F

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$$\sin \alpha = \frac{2}{4}$$

M1

$$\alpha = \frac{\pi}{6}$$

A1

$$\text{Range is } \frac{\pi}{3} \leq \arg z \leq \frac{2\pi}{3}$$

Deduct 1 for angles in degrees

A1

4

[8]

Q12.

- (a) Centre $-1 - i$ or $(-1, -1)$

B1

Radius 5

ft incorrect centre if used

M1A1F

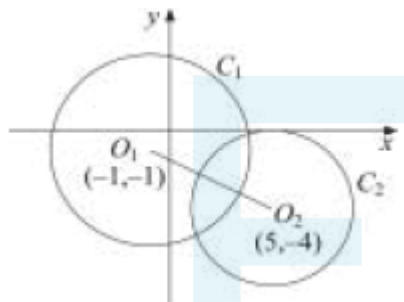
$$|z + 1 + i| = 5 \text{ or } |z - (-1 - i)| = 5$$

ft $|z + 1 + i| = 10$ earns M0B1

A1F

4

(b)



C_1 correct centre, correct radius

ft errors in (a) but fit circles need to intersect and C_1 enclose $(0, 0)$

B1F

C_2 correct centre, correct radius

B1

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Touching x-axis

error in plotting centre

B1F

3

(c) $O_1O_2 = 3\sqrt{5}$

allow if circles misplaced but

O_1O_2 is still $3\sqrt{5}$

M1A1

Correct length identified

m1M1

Length is $9 + 3\sqrt{5}$

ft if r is taken as 10

A1F

Q13.

(a) (i) $z = -i \quad \left| \frac{-2\sqrt{3} - 2i}{-2\sqrt{3} - 2i} \right| = \frac{\sqrt{12 + 4}}{\sqrt{12 + 4}} = 4$

4

M1

A1

2

(ii) Centre of circle is $2\sqrt{3} + i$
Do not accept $(2\sqrt{3}, 1)$ unless attempt to solve using trig

Substitute into line

B1

$\arg(2\sqrt{3} + 2i) = \frac{\pi}{6}$ shown

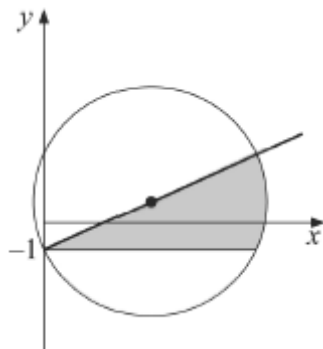
M1

A1

3

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(b)



Circle: centre correct

B1

through $(0, -1)$

B1

Half line: through $(0, -1)$

B1

through centre of circle

B1

4

- (c) Shading inside circle and below line

B1F

Bounded by $y = -1$

B1

2

[11]

Q14.

- (a) radius $\sqrt{2}$ centre $-5 + i$
condone $(-5, 1)$ for centre do not accept $(-5, i)$

B1,B1

2

- (b) $\arg(z_1 + 2i) = \arg(-4 + 4i)$

M1

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clearly shown eg $\tan^{-1}\left(\frac{1}{1}\right)$

A1

2

- (c) (i) $|z_1 + 5 - i| = |1 + i| = \sqrt{2}$

B1

1

- (ii) Gradient of line from
 $(-5, 1)$ to $(-4, 2)$ is 1 $\left(\frac{\pi}{4}\right)$

M1 for a complete method

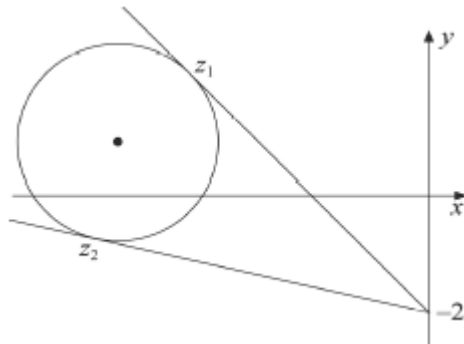
M1A1

radius $\perp$ line $\therefore$ tangent

E1

3

(iii)



Circle correct

ft incorrect centre or radius

B1F

Half line correct

line must touch C generally above the circle

B1

2

(d) z_2 in correct place

B0 if z_2 is directly below the centre of C

B1

with tangent shown

B1

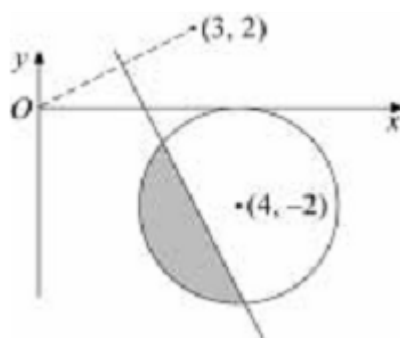
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[12]

Q15.

(a)



(i)

Circle

Correct centre

B1

Correct radius

B1

Touching x-axis

B1

3

(ii) **Line**

Point (3, 2) indicated

B1

Line through $\left(1\frac{1}{2}, 1\right)$

B1 ✓

Perpendicular to $(0, 0) \rightarrow (3, 2)$

B1

3

(b) Correct shaded area

For shading inside the circle provided no other area is shaded

B1

Must be a circle and a straight line for second B1

B1 ✓

2

[8]

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Q16.

(a) Explanation

E1 for $i = e^{\frac{\pi i}{2}}$ or $iz_1 = -y_1 + ix_1$

E2,1,0

2

(b) (i) Perpendicular bisector of AB

B1

through O

B1

2

(ii) half-line

If L_2 is taken to be the line AB give B0

B1

from B

B1

parallel to OA

B1

3

(c) $(1 + i)z_1$

ft if L_2 taken as line AB

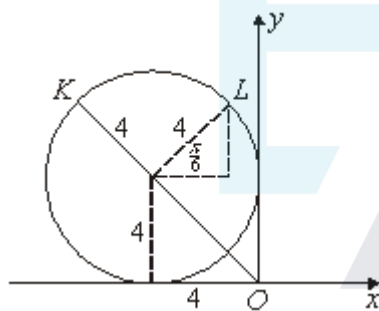
M1A1

2

[9]

Q17.

(a)



Circle

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Correct centre

B1

Touching both axes

B1

B1

3

(b) $|z| \text{ max} = OK$

Accept $\sqrt{4^2 + 4^4} + 4$ as a method

M1

$$= \sqrt{4^2 + 4^4} + 4$$

Follow through circle in incorrect position

A1F

$$= 4(\sqrt{2} + 1)$$

AG

A1F

3

(c) Correct position of z , ie L

M1

$$a = -\left(4 - 4 \cos \frac{1}{6} \pi\right)$$

$$= -(4 - 2\sqrt{3})$$

Follow through circle in incorrect position

A1F

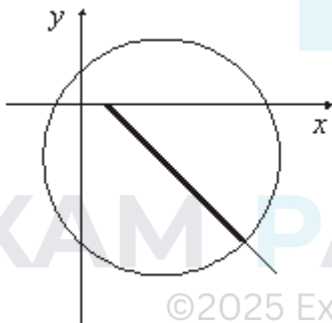
$$b = 4 + 4 \sin \frac{1}{6} \pi = 6$$

A1F

3

[9]

Q18.



(a) (i) Circle

B1

Correct centre

B1

Enclosing the origin

B1

3

(ii) Half line

B1

Correct starting point

B1

Correct angle

B1 3

(b) Correct part of the line **indicated**

B1F 1

[7]



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Examiner reports

Q1.

Part (a) was well answered, although most circles were drawn freehand. However, an accurately drawn circle made both (b)(i) and (b)(ii) easier to solve.

Most students realised that the required line was a tangent to the circle, but many of these failed to realise that a simple sine ratio could be used to find the angle α .

Part (b)(ii) was only answered well by only the most able students. Although most of the successful attempts used basic GCSE trigonometry, some elaborate methods were also employed, including the solution of quadratic simultaneous equations. A common error was to assume that the triangle with vertices 0, 3 and w was right-angled. Another common mistake was to assume that θ and α were equal.

Q3.

- (a) Very few students scored full marks for the verification. It was not sufficient to merely write $|4 - 4i| = 4\sqrt{2}$; some indication as to how the modulus was being found needed to be shown.

Similarly “ $\arg(-2 + 2i) = -\frac{\pi}{4} = \frac{3\pi}{4}$ ”, which was presented as a solution by many, was not deemed worthy of a mark for verification. A sketch with a calculation often convinced examiners that the argument had the correct value.

- (b) It was pleasing to see how well students understood these basic loci, and, although some could have taken greater care drawing a circle, a freehand sketch was acceptable. The main error was in drawing the half line where quite a few did not realise that it started at $0 - i$ on the Argand diagram.
- (c) Those who realised that the half line passed through the centre of the circle had little trouble in finding the complex number represented by Q . Another successful approach involved forming solving a quadratic equation based on the Cartesian equations of the circle and line. A few did the hard part of the question but wrote their final answer as coordinates and so only scored the method mark.

Q4.

- (a) Almost everyone managed to draw a circle for the locus with the vast majority realizing that the centre was at $6i$. Some drew a circle with its centre below the real axis and a few thought that the centre lay on the real axis. It was necessary to indicate that the radius of the circle was 3 and most did this by showing the values where the circle cut the imaginary axis. Others indicated the radius of 3 in a variety of ways but quite a few neglected to make this clear to examiners.
- (b) (i) Several students wrote $9i$ instead of 9 as the maximum value of the modulus of z . It was evident that some had no idea what the question was really asking.
- (ii) Most students realised the need to draw a tangent to the circle but weaker students could not cope with the basic trigonometry required to find a relevant angle on their diagram. Nevertheless a large number had no problem in finding the correct maximum value of the argument of z .

Q5.

Whilst part (a) was drawn correctly by almost all candidates, the reverse was true in part (b). In this part virtually all candidates drew their circle touching $\text{Im}(z) = 0$ instead of the $\text{Re}(z) = 0$, clearly not understanding the statement $\text{Re}(z) = 0$. This caused a significant loss of marks, especially as the answer to part (b)(iii), the least value of $\arg z_1$, became trivial if a candidate's circle did not touch $\text{Re}(z) = 0$. Those candidates whose circle did not touch $\text{Im}(z) = 0$, usually drew their circle through the origin, and this also precluded them from any viable attempt at part (b)(iii).

Q6.

It was pleasing to see many neat and fully or nearly correct answers to this question. It was clear that students had taken considerable care over their diagrams. The circle was well drawn, but the perpendicular bisector perhaps not quite so well. A few attempts at the perpendicular bisector were drawn through the origin, whilst occasionally others were drawn parallel to one of the axes. The final mark was lost by some students who shaded the minor segment between the line and the circle, rather than identify the minor arc of the circle.

Q7.

Part (a) was well done and almost universally correct. However part (b) proved to be beyond some of the most able candidates. Candidates were just not able to identify the correct position of P , the commonest positions being either at the origin or at the end of the diameter of the circle parallel to the x -axis. Even when the point P was correctly identified, not all were able to use simple geometrical properties to find the length of OP .

Q8.

The most satisfactory solutions came from candidates who were careful with their drawings and who used the same scale on both their axes. In part (a)(i), marks were generally lost by candidates either misplotting the centre of the circle, or by failing to realise that the circle touched the Real axis. In part (a)(ii), a number of candidates drew their half line from the origin and even when it did start at the point $(0, -2)$, it failed to pass through the point of contact of the circle with the Real axis. Part (b) was not particularly well done as many candidates shaded a region within the circle or on occasions marked just the end points of the chord rather than the full chord of the circle.

Q9.

This question was well done overall and many candidates scored all of the eight available marks. Those candidates using mathematical instruments, ie ruler and compasses, produced superior solutions. Errors, when they did occur, were either the misplotting of the centre of the circle, or more commonly, the misplotting of the line. The commonest mistakes were either to draw the line through the point $(0, 2)$ or more frequently through $(0, -1)$. If serious errors were made in the plotting of the line, loss of marks in the shading were almost inevitable.

Q10.

The verifications in part (a) were not always convincing, especially the verification that the point representing the complex number $2 + 2i$ lay on the line L_1 . The sketches in part (b) varied considerably. Those candidates who made a reasonably careful drawing generally scored higher marks as they were able to clearly show that the point representing $2 + 2i$

lay on both L_1 and L_2 . Careful sketches also improved a candidate's chance of scoring full marks in part (c).

Q11.

Part (a) was well done apart from a few candidates who drew their circle at the mirror image of its correct position in the x -axis. In part (b), although candidates realised that the tangents needed to be drawn from the origin to the circle in order to find the possible value of $\arg z$, few were able to manage the trigonometry involved to reach the correct

range, and it was not uncommon to see $\tan^{-1} \left(\frac{4}{2} \right)$ appearing, suggesting that candidates thought that these points were in fact the points of intersection of the circle with the line through its centre parallel to the x -axis.

Q12.

The coordinates of the centre of the circle in part (a) were usually obtained, but the notation was often poor and it was not uncommon to see the centre of the circle C_1 written as $(-1, -i)$ and, on the diagram, the scale on the y -axis written as $i, 2i, 3i$ and so on. Also, radius and diameter were commonly confused.

Sketches in part (b) varied considerably, the best being those who used compasses for their circles. These were generally readable with centre and radius indicated. However, some candidates chose to draw their circles by plotting points and joining up by freehand. These sketches turned out to be very poor. The circle C_2 was sometimes mistakenly drawn in the incorrect quadrant through choice of centre as $(-5, 4)$ whilst others either failed to realise that the circle C_2 touched the x -axis or drew a circle touching both axes.

In part (c), although many candidates placed z_1 and z_2 on their diagram in the approximately correct positions, not all realised that these points were at the intersections of C_1 and C_2 and the line O_1O_2 , and even when they did, the finding of the length O_1O_2 proved to be beyond many.

Q13.

Lack of clear evidence that candidates understood what they were doing in part (a) caused a loss of marks for this part of the question. Methods varied. Those candidates who turned this part of the question into a coordinate geometry exercise probably provided the clearest solutions.

Those candidates who evaluated $|-2i - 2\sqrt{3}|$ and $\arg(2\sqrt{3} + 2i)$ provided less convincing solutions and in some cases evident error. For instance it was not uncommon to see $|-2i - 2\sqrt{3}|$ written as $\sqrt{(2\sqrt{3})^2 - (2i)^2}$. Part (b) on the whole was done well except that in some instances not all the results of part (a) were incorporated in the candidate's Argand diagram. Part (c) was done well but, if a mistake did occur, it was almost always that the shaded area would be bounded by the real axis rather than by a line parallel to the real axis through the point represented by the complex number $z = -i$.

Q14.

Apart from the odd sign error, part (a) was done correctly. Answers to part (b) and to

some extent part (c)(i) lacked detail especially as the results were given or implied. Very few candidates scored credit in part (c)(ii) and when it was attempted it was almost always solved by finding the cartesian equation of the circle C and the line L and by showing that the resulting quadratic equation giving the points of intersection of C and L had two coincident roots. This method, of course, involved a considerable amount of algebraic manipulation, when in fact the consideration of the gradient of the line joining the centre of C to the point represented by z_1 would have given the result immediately. However, this was very rarely seen. It was surprising to see many candidates draw the circle C and the line L intersecting in two points on the Argand diagram (part (c)(iii)) in spite of the results candidates had been requested to show in the earlier parts of the question. There were many incorrect solutions to part (d), the most common being to place z_2 vertically below the centre of C .

Q15.

A few candidates misplotted the centre of the circle, usually at $(-4, 2)$. Apart from this most drew the circle correctly. Not all recognised the line as the perpendicular bisector of the line joining the origin to the point $(3, 2)$ in part (b), but rather thought that this equation represented another circle. This in turn had an effect on the shading in part (c) although the interior of the circle was usually shaded. It should be said that the diagrams were neat and in the main well labelled and in proportion; a great improvement on sketches submitted in previous years.

Q16.

Explanations in part (a) were very unclear and generally far from convincing. Candidates generally referred to what had happened to the coordinates of the points represented by z_1 and z_2 , but few made allusion to the significance of i in the iz_1 . The neatest solutions came from candidates who considered multiplication of a complex number by i as a rotation anticlockwise of $1/2\pi$. Inaccurate copying of the diagram in part (b) caused loss of marks. For instance, although candidates knew that the locus L_1 was the perpendicular bisector of AB , poor diagrams meant that their line did not pass through the origin. Again, for the locus L_2 , although the majority of candidates drew a half line through B , their line was not always parallel to OA . Part (c) proved to be beyond most candidates probably because few realised that the point of intersection of L_1 and L was, in fact, the fourth vertex of the square whose three other vertices were A , O and B .

Q17.

Most candidates realised that the locus in part (a) was a circle although it was frequently drawn in an incorrect quadrant, and occasionally with a radius of 2 rather than a radius of 4. The correct answer to part (b) was usually obtained although sometimes with a less than convincing argument. There were relatively fewer correct solutions to part (c). Those candidates who addressed the geometry of the figure were the most successful, but those who converted the equations of the circle and line into the Cartesian form and then attempted to solve a pair of simultaneous equations usually abandoned their solution after making algebraic errors.

Q18.

Responses to this question were usually quite good and it was pleasing to note some quite accurate neat diagrams using a ruler and compasses. Errors in parts (a)(i) and part (a)(ii) were usually errors of sign. For instance in part (a)(i) the centre of the circle was

sometimes taken to be the point $(-3,2)$ or even $(3,2)$ and in part (a)(ii) the half line would be drawn from either $(0,1)$ or $(-1,0)$. Just occasionally the radius of the circle was taken to

be 2, or the direction of the line was taken to be $+\frac{\pi}{4}$ or $+\frac{3\pi}{4}$. In part (b), a substantial number of candidates thought that the set of points must involve an area and consequently shaded some region in their sketch.



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